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ISOCELL

The **ISOCELL** CMOS camera sensors are a family of sensors produced by Samsung and available for purchase by other companies. They are used in a wide variety of products including mobile phones, computers and digital cameras.[1]

Design

These sensors use one of the following pixel type technologies:

- **FSI**: frontside-illuminated. The light that reaches the photosensitive area is reduced because it needs to pass through multiple metal and dielectric layers.
- **BSI**: backside-illuminated.The metal wiring shifted to backside of photosensitive area(photodiode). The light reaches the photosensitive area directly.
- **ISOCELL**: ISOCELL combines 3D-BSI with Front-side, Full-depth, Deep-Trench Isolation (F-DTI) and Vertical Transfer Gate (VTG). This provides increased light sensitivity and higher color fidelity even in poor lighting conditions.[2]
- **ISOCELL Plus**: ISOCELL Plus replaces ISOCELL's metal grid barriers with an innovative new material developed by Fujifilm. This minimizes optical loss and light reflection, providing higher color fidelity and up to a 15% enhancement in light sensitivity compared to ISOCELL.[3]
- **ISOCELL 2.0**: ISOCELL 2.0 builds on ISOCELL Plus technology by additionally replacing the lower portion of the color filter barriers with a more reflective material. It further reduces optical loss in each pixel and drastically improves light sensitivity producing even more vivid pictures with reduced noise.[4]

These sensors can be configured with one of the following chroma technologies:

- **BW** also known as Monochrome. No color filter array.
- **RGB** also known as Bayer filter or RGGB. Features a repeating 2×2 pattern with 1 red, 2 green and 1 blue pixels.
- **Tetracell** also known as Quad Bayer or 4-cell. For darker scenes, signal processing can combine data from 2x2 pixel groups to essentially act like a larger pixel with a repeating 4×4 subpixel pattern with 4 red, 8 green and 4 blue subpixels. For brighter scenes, signal processing can convert the Tetracell into a conventional RGB filter to achieve 4x higher resolution. Resolution & pixel size listed below is after re-mosaic signal processing.[5]
- **Nonacell** is similar to Tetracell, but with 3x3 pixel groups and a 6×6 pattern with 9 red, 18 green and 9 blue subpixels.
- **ChameleonCell** is similar to Tetracell and Nonacell, but with 4x4 pixel groups and a 8x8 pattern with 16 red, 32 green and 16 blue subpixels.

List of sensors

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL KD1 (S5KKD1)[6]	6560 x 4928 32 MP	10	8/10	1/3.42" (5.25 mm)	0.64	ISOCELL Plus	Tet Bay
ISOCELL GN8 (S5KGN8)[7]	8192 x 6144 50 Mp	10	10/12	1/1.95" (8.19 mm)	0.8	ISOCELL Plus	Tet Bay
ISOCELL HP9 (S5KHP9)[8] 2024.06.27	16320 x 12288 200 MP 8K@30 fps 4K@120 fps FHD@480 fps	10	10/12/14	1/1.4"	0.56	ISOCELL 3.0	Tet RG
ISOCELL GNJ (S5KGNJ)[9] 2024.06.27	8192 x 6144 50 MP 8K@30 fps 4K@120 fps FHD@480 fps	10	8/10/12/14	1/1.56"	1.0	ISOCELL 3.0	Du RG
ISOCELL JN5 (S5KJN5)[10] 2024.06.27	8192 x 6144 50 MP 4K@60 fps FHD@240 fps	10	10	1/2.76"	0.64	ISOCELL 3.0	Tet Bay
ISOCELL GNK (S5KGNK)	8160 x 6144 50 MP Video: 8K@30 fps 4K@120 fps FHD@240 fps	10	8/10/12/14	1/1.3"	1.2	ISOCELL 3.0	Du RG
ISOCELL HP2 (S5KHP2)[11] 2023.01.17	16384 x 12288 200 MP Video: 8K@30 fps 4K@120 fps	10	10/12	1/1.3" (12.31 mm)	0.6	ISOCELL 3.0	Tet RG
ISOCELL HPX 2022.10.20	16320 x 12288 200 MP Video: 8K@30 fps 4K@120 fps	10	10/12/14	1/1.4" (11.43 mm)	0.56	ISOCELL 3.0	Tet RG
ISOCELL HP3 (S5KHP3)[12] 2022.06.23	16320 x 12288 200 MP Video: 8K@30 fps 4K@120 fps	10	10/12/14	1/1.4" (11.43 mm)	0.56	ISOCELL 3.0	Tet RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL HP1 (S5KHP1)[13] 2021.09.02	16384 x 12288 200 Mp Video: 8K@30 fps	10	8/10/12	1/1.22" (13.11 mm)	0.64	ISOCELL 3.0	Tet RG
ISOCELL HM6 (S5KHM6)[14] 2022.03.	12000 x 9000 108 Mp Video: 8K@24 fps	10	8/10/12	1/1.67" (9.58 mm)	0.64	ISOCELL 2.0	Nor RG
ISOCELL HM3 (S5KHM3)[15] 2021.01.14	12000 x 9000 108 Mp Video: 8K@30 fps	10	8/10/12	1/1.33" (12.03 mm)	0.8	ISOCELL Plus	Nor RG
ISOCELL HM2 (S5KHM2)[16] / 2020.09.15	12000 x 9000 108 Mp Video: 8K@24 fps	10	8/10	1/1.52" (10.53 mm)	0.7	ISOCELL Plus	Nor RG
ISOCELL HMX (S5KHMX)[17] / 2019.08.12	12032 x 9024 108 Mp Video: 6016 x 3384 @ 30 fps	10	10	1/1.33" (12.03 mm)	0.8	ISOCELL Plus	Tet Bay
ISOCELL HM1 (S5KHM1)[18] / 2020.02.12	12000 x 9000 108 Mp Video: 8K@24 fps	10	8/10	1/1.33" (12.03 mm)	0.8	ISOCELL Plus	Nor RG
ISOCELL GW3 (S5KGW3)[19] / 2020.09.15	9280 x 6944 64 Mp Video: 3840 x 2160 @ 60 fps	10	10	1/1.97" (8.12 mm)	0.7	ISOCELL Plus	Tet Bay
ISOCELL GW2 (S5KGW2)[20] / 2020.	9280 x 6944 64 Mp Video: 7680 x 4320 @ 24 fps	10	8/10	1/1.72" (9.21 mm)	0.8	ISOCELL Plus	RG
ISOCELL GW1 (S5KGW1)[21] / 2019.05.09	9280 x 6944 64 Mp Video: 7680 x 4320 @ 24 fps	10	8/10	1/1.72" (9.21 mm)	0.8	ISOCELL Plus	Tet Bay

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL GN9 (S5KGN9)[22] / 2023	8160 x 6144 50 Mp	10	10	1/1.57" (10.21 mm)	1.0	ISOCELL Plus	Tet Ba
ISOCELL GN5 (S5KGN5)[23] / 2021.09.02	8160 x 6144 50 Mp Video: 7680 x 4320 @ 30 fps	10	8/10/12	1/1.57" (10.19 mm)	1.0	ISOCELL 2.0	Du RG
ISOCELL GN3 (S5KGN3)[24] 2022.08.13	8160 x 6144 50 Mp Video: 7680 x 4320 @ 30 fps	10	8/10/12	1/1.57" (10.19 mm)	1.0	ISOCELL 2.0	Du RG
ISOCELL GN2 (S5KGN2)[25] / 2021.02.23	8160 x 6144 50 Mp Video: 7680 x 4320 @ 24 fps	10	8/10	1/1.12" (14.28 mm)	1.4	ISOCELL Plus	Du RG
ISOCELL GN1 (S5KGN1)[26] / 2020.05.19	8160 x 6144 50 Mp Video: 7680 x 4320 @ 30 fps	10	8/10	1/1.31" (12.21 mm)	1.2	ISOCELL Plus	Du RG
ISOCELL JN1 (S5KJN1)[27] / 2021.06.10	8160 x 6144 50 Mp Video: 3840 x 2160 @ 60 fps	10	10	1/2.76" (6.54 mm)	0.64	ISOCELL 2.0	Tet Ba
ISOCELL GM5 (S5KGM5)[28] / 2020.09.15	8064 x 6048 48 Mp Video: 7680 x 4320 @ 30 fps	10	8/10	1/2.55" (7.06 mm)	0.7	ISOCELL 2.0	Tet Ba
ISOCELL GM2 (S5KGM2)[29] / 2019.05.09	8000 x 6000 48 Mp	10	8/10	1/2.0" (8 mm)	0.8	ISOCELL Plus	Tet Ba

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
	Video: 3840 x 2160 @ 60 fps						
ISOCELL GM1 (S5KGM1)[30] / 2018.10.30	8000 x 6000 48 Mp Video: 3840 x 2160 @ 60 fps	10	8/10	1/2.0" (8 mm)	0.8	ISOCELL Plus	Tet Bay
ISOCELL GH1 (S5KGH1)[32] / 2019.09.24	7968 x 5480 43.7 Mp Video: 3840 x 2160 @ 60 fps	10	10	1/2.65" (6.77 mm)	0.7	ISOCELL Plus	Tet Bay

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL JD1 (S5KJD1)[33] / 2020.09.15	6560 x 4920 32 Mp	10	10	1/3.14" (5.1 mm)	0.7	ISOCELL Plus	Tet Bay
ISOCELL GD2 (S5KGD2)	6560 x 4928 32 Mp	10	8/10	1/2.8"	0.8	ISOCELL Plus	Tet Bay
ISOCELL GD1 (S5KGD1) / 2018.10.30	6560 x 4928 32 Mp	10	8/10	1/2.80" (6.4 mm)	0.8	ISOCELL Plus	Tet Bay
ISOCELL 2L3 (SAK2L3)[34]	4032 x 3024 12.2 Mp Video: 3840 x 2160 @ 60 fps	10	8/10	1/2.55" (7.06 mm)	1.4	ISOCELL	RG
ISOCELL 2L1 (S5K2L1)[35]	4032 x 3024 12.2 Mp	10	8/10	1/2.6" (7.06 mm)	1.4	ISOCELL	RG
ISOCELL 2L2 (S5K2L2)[37]	4032 x 3024 12.2 Mp	10	8/10	1/2.55" (7.06 mm)	1.4	ISOCELL	RG
ISOCELL 2L3 (S5K2L3)[38]	4032 x 3024 12.2 Mp Video: 3840 x 2160 @ 60 fps			1/2.55" (7.06 mm)	1.4	ISOCELL Plus	RG
ISOCELL 2L4 (S5K2L4)[40]	4032 x 3024 12.2 Mp Video: 3840 x 2160 @ 60 fps	10	8/10	1/2.55" (7.06 mm)	1.4	ISOCELL Plus	RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL 2L7 (S5K2L7)[42]	4032 x 3024 12.2 Mp	10	8/10	1/2.56" (7.06 mm)	1.4	ISOCELL	RG
ISOCELL 2L8 (S5K2L8)[45]	4032 x 3024 12.2 Mp	10	8/10	1/2.80" (6.4 mm)	1.28	ISOCELL	RG
ISOCELL 2L9 (S5K2L9)	4032 x 3024 12.2 Mp			1/2.55" (7.06 mm)	1.4	ISOCELL	RG
ISOCELL 2LD (S5K2LD)	4032 x 3024 12.2 Mp Video: 3840 x 2160 @ 60 fps			1/1.76" (9.072 mm)	1.8	ISOCELL Plus	RG
ISOCELL 2M8 (S5K2M8)	4208 x 3120 13 Mp			1/2.60" (6.828 mm)	1.34	ISOCELL	RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (µm)	Pixel type	Co
ISOCELL 2P2 (S5K2P2)	5312 x 2988 16 Mp			1/2.60" (6.828 mm)	1.12	ISOCELL	RG
ISOCELL 2P7 (S5K2P7)[49]	4640 x 3488 16 Mp	10	8/10	1/2.80"	1.12	ISOCELL	RG
ISOCELL 2P8 (S5K2P8)	5336 x 3000 16 Mp			1/2.60" (6.828 mm)	1.12	ISOCELL	RG
ISOCELL 2T8 (S5K2T8)	5976 x 3368 20 Mp			1/2.60" (6.828 mm)	1.1	ISOCELL	RG
ISOCELL 3H1 (S5K3H1)	3280 x 2464 8 Mp			1/3.60"	1.22	ISOCELL	RG
ISOCELL 3H2 (S5K3H2)	3264 x 2448 8 Mp			1/3.20" (5.68 mm)	1.4	ISOCELL	RG
ISOCELL 3J1 (S5K3J1)[51]	3648 x 2736 10 Mp			1/3.20" (5.57 mm)	1.22	ISOCELL Plus	RG
ISOCELL 3L2 (S5K3L2)[53]	4208 x 3120 13 Mp			1/3.06" (5.867 mm)	1.12	ISOCELL	RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL 3L8 (S5K3L8)[55]	4208 x 3120 13 Mp			1/3.06" (5.867 mm)	1.12	ISOCELL	RG
ISOCELL 3M2 (S5K3M2)	4208 x 3120 13 Mp			1/3.06" (5.867 mm)	1.12	ISOCELL	RG
ISOCELL 3M3 (S5K3M3 & S5K3M3+)	4208 x 3120 13 Mp	10	8/10	1/3.4"	1.0	ISOCELL	RG
ISOCELL 3M5 (S5K3M5)[59]	4208 x 3120 13 Mp			1/3.40"	1.0	ISOCELL	RG
ISOCELL 3P3 (S5K3P3)	5312 x 2988 16 Mp			1/3.06" (5.867 mm)	1.0	ISOCELL	RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (µm)	Pixel type	Co
ISOCELL 3P8 (S5K3P8)[64]	4640 x 3488 16 Mp			1/3.10" (6.4 mm)	1.0	ISOCELL	RG
ISOCELL 3P9 (S5K3P9)[65]	4608 x 3456 16 Mp			1/3.10" (6.4 mm)	1.0	ISOCELL	RG
ISOCELL 3T1 (S5K3T1SP - Tetrapixel, S5K3T1SX - RGB)	5184 x 3880 20 Mp	10	8/10	1/3.0" (6.4 mm)	0.9	ISOCELL	Tet

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (μm)	Pixel type	Co
ISOCELL 3T2 (S5K3T2)[66]	5184 x 3880 20 Mp			1/3.40"	0.8	ISOCELL Plus	Tet
ISOCELL 4E6 (S5K4E6)	2592 x 1944 5 Mp			1/4" (4.5 mm)	1.34	ISOCELL	RG
ISOCELL 4E8 (S5K4E8)	2592 x 1944 5 Mp			1/4" (4.5 mm)	1.4	ISOCELL Bright	RG
ISOCELL 4H5 (S5K4H5)	3264 x 2448 8 Mp			1/4" (4.5 mm)	1.12	BSI(~YB)/ ISOCELL(~YC)	RG
ISOCELL 4H7 (S5K4H7)[69]	3264 x 2448 8 Mp			1/4" (4.5 mm)	1.12	BSI	RG
S5K5BAF[70]	1600 x 1200			1/5"	1.75	FSI	RG

Model (Part Number) Release date	Resolution	ADC accuracy (bits)	Output formats RAW (bits)	Diagonal in Optical format (or mm)	Pixel size (µm)	Pixel type	Co
S5K5E2	2560 x 1920 5 Mp			1/5" (5.08 mm)	1.12	BSI	RG
S5K5E3	2576 x 1932 5 Mp			1/4" (4.5 mm)	1.12	BSI	RG
S5K5E8	2592 x 1944 5 Mp			1/5" (5.08 mm)	1.12	BSI	RG
S5K6A3[72]	1412 x 1412			1/6" (3 mm)	1.75	BSI	RG
S5K6B2	1936 x 1036 2 Mp			1/6" (3 mm)	1.34	BSI	RG
S5K6D1	2560 x 1440 3.7 Mp			1/5" (3.6 mm)	1.12	BSI	RG

[1]: The Galaxy S7 family utilizes either one of two sensors: the Samsung S5K2L1 or the Sony IMX260. They are regarded as nearly identical sensors.

[2]: The Galaxy S6 family and Note 5 utilizes either one of two sensors: the Samsung S5K2P2 or the Sony IMX240. They are regarded as nearly identical sensors.

[3]: The Xiaomi Mi 4C and Oneplus X utilize either one of two sensors: the Samsung S5K3M2 or the Sony IMX258. They are similar sensors, but the IMX258 is a newer unit.

[4]: The Galaxy S8 family utilizes either one of two sensors: the Samsung S5K2L2 or the Sony IMX333. They are regarded as nearly identical sensors.

[5]: The Xiaomi Mi 6 has two sensors. One of the two sensors is S5K3M3, another is Sony IMX386.

[6]: The Meizu Pro 6 family utilizes either one of two sensors: the Samsung S5K3M3 or the Sony IMX386, They are regarded as nearly identical sensors

[7]: The Xiaomi Mi Max 2 utilizes either one of two sensors: the Samsung S5K3M3 or the Sony IMX386. They are regarded as nearly identical sensors

[8]: The Samsung Galaxy S9 and S9+ utilize either one of two sensors: the Samsung S5K2L3 or the Sony IMX345. Both feature stacked LPDDR4 DRAM in the CMOS sensor.

[9]: The Galaxy S8 family utilizes either one of two sensors: the Samsung S5K3H1 or the Sony IMX320. They are regarded as nearly identical sensors.

[10]: The Galaxy S10 family utilizes either one of two sensors: the Samsung S5K3J1 or the Sony IMX374. They are regarded as nearly identical sensors.

[11]: The Galaxy Z Flip's ultra-wide camera utilizes the 16 Mp sensor, but images are cropped down to 12 Mp.

[12]: The Galaxy S21 and S21+ utilizes either one of two sensors: the Samsung 2LD (Exynos variant) or the Sony IMX555 (Snapdragon variant).[76] They are regarded as nearly identical sensors.

[13]: The backlight sensor is a special photographic sensor. It allows you to take high quality, clear photos in low light. The BSI sensor improves image sharpness during shooting.[77]

See also

- Exmor
- Bionz – image processor
- HAD CCD – Sony
- Expeed – Nikon image/video processors
- Toshiba CMOS
- OmniVision

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